

SECTION 5

PERFORMANCE DATA

TABLE OF CONTENTS

	<u>Page</u>
5.1 GENERAL	5-1
5.1.1 Definitions	5-1
5.1.1.1 Airspeeds	5-1
5.1.1.2 Height	5-1
5.1.1.3 Outside Air Temperature (OAT)	5-2
5.1.1.4 Pressure Altitude	5-2
5.1.2 Standard Performance Conditions	5-2
5.1.3 Conditions Affecting Performance	5-2
5.1.4 Reading the Charts	5-2
5.2 INFLIGHT POWER CHECKS	5-3
5.2.1 Power Assurance Check	5-3
5.2.2 Engine Trend Check	5-8
5.2.3 Power Trend Monitoring	5-8A
5.3 DENSITY ALTITUDE	5-8B
5.4 AIRSPEED CALIBRATION	5-10
5.5 HEIGHT-VELOCITY DIAGRAMS	5-12
5.6 HOVER CEILING	5-16
5.7 RATE OF CLIMB	5-21
5.8 CLEAR AIRFIELD LANDING DISTANCE	5-33

LIST OF FIGURES

	<u>Page</u>
Fig. 5-1 Inflight Power Check (Anti-icing - OFF)	5 - 5
Fig. 5-1A Inflight Power Check (Anti-icing - ON)	5 - 7
Fig. 5-2 Plotting Example for Power Trend Monitoring	5 - 8A
Fig. 5-3 Density Altitude Chart	5 - 9
Fig. 5-4 Airspeed Calibration Chart	5 - 11
Fig. 5-5 Height-velocity Diagrams - Single Engine Failure	5 - 13
Fig. 5-6 Height-velocity Diagrams - Double Engine Failure	5 - 14
Fig. 5-7 Hover Ceiling I G E - TOP	5 - 17
Fig. 5-8 Hover Ceiling I G E - MCP	5 - 18
Fig. 5-9 Hover Ceiling O G E - TOP	5 - 19
Fig. 5-10 Hover Ceiling O G E - MCP	5 - 20
Fig. 5-11 Rate of Climb - AEO, TOP	5 - 23
Fig. 5-12 Rate of Climb - AEO, MCP	5 - 27
Fig. 5-13 Rate of Climb - OEI, EPWR	5 - 31
Fig. 5-14 Rate of Climb - OEI, MCP	5 - 32
Fig. 5-15 Landing Profile - Clear Airfield	5 - 33
Fig. 5-16 Landing Distance - Clear Airfield	5 - 35

SECTION 5

PERFORMANCE DATA

5.1 GENERAL

This section contains the performance information necessary for preflight and inflight planning. The information is primarily intended for mission planning and is most useful when planning operations in unfamiliar areas or at extreme conditions. Although maximum performance is not always required, regular use of the information in this section is recommended for the following reasons:

- Knowledge of performance margins will allow better decisions when unexpected conditions or alternate missions are encountered.
- Situations requiring maximum performance will be more readily recognized.
- Familiarity with the data will allow performance to be calculated more easily and quickly.
- Experience will be gained in accurately estimating the effects of conditions for which data is not presented.

5.1.1 Definitions

Some terms which are used particularly in the performance section are defined as follows:

5.1.1.1 Airspeeds

IAS The indicated airspeed is the reading obtained on a pitot-static airspeed indicator having no instrument calibration error. If the calibration error on the particular instrument is not known, then the actual reading may be taken equal to IAS because the tolerances permitted on the instrument are small.

CAS The calibrated airspeed is the IAS-value corrected for pitot-static position errors only.

TAS The true airspeed of the helicopter relative to undisturbed air. It can be obtained by multiplying the CAS value by the true airspeed factor, if compressibility is neglected.

V_{NE} The never exceed speed, IAS.

V_Y The speed for best rate of climb, IAS.

5.1.1.2 Height

The true vertical clearance between the lowest part of the helicopter and the relevant datum.

5.1.1.3 Outside Air Temperature (OAT)

The ambient temperature of the free airstream, expressed in °C.

5.1.1.4 Pressure Altitude

The expression of atmospheric pressure in terms of altitude, according to the interrelation of these factors in the international standard atmosphere. Pressure altitude may be obtained by setting the subscale of an accurate barometric altimeter to 1013.2 hPa (29.92 inches or 760 mm of mercury).

NOTE All altitudes given in this manual are pressure altitudes, unless otherwise indicated.

5.1.2 Standard Performance Conditions

The data presented cover the maximum range of performance that can be expected and are based on the following conditions:

- Engine power does not exceed limits (see Section 2).
- Installation and accessory losses are included.
- Rotor RPM - 100 %.

In each area of performance, the effects of altitude, temperature, gross mass, and other parameters relating to that phase of flight are presented.

5.1.3 Conditions Affecting Performance

Conditions affecting performance are listed in the heading of the respective charts. The data presented are accurate only for these specific conditions.

Other general conditions applicable to the performance data are:

- Airframe and engine control rigging (assumed within allowable tolerances).
- Pilot technique (assumed normal, control movements smooth and continuous).
- Clean performance influencing components (rotor blades, engine air intakes and compressor sections).

5.1.4 Reading the Charts

None of the curves presented should be extrapolated, but interpolation between given data is permissible.

5.2 INFLIGHT POWER CHECKS

The power assurance check is provided to ensure that the engines are capable of developing the power used in establishing the performance data presented in this manual.

The engine trend check is provided to periodically monitor the long-term engine behavior.

5.2.1 Power Assurance Check

This check is performed in cruise flight with one engine operating at IDLE; each engine being checked separately, in turn. The generator and non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

The power assurance check must be performed at intervals not to exceed 100 flight hours or whenever abnormal engine function is noted.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).

WARNING MAINTAIN A SUITABLE MINIMUM SAFE HEIGHT IN CASE OF ENGINE FAILURE

2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y).

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical load as much as possible.
5. Switch off generator of the engine to be checked.
6. Set power lever of non-check engine slowly to IDLE.

NOTE Observe limits (N_1 , TOT, Torque) of check engine.

7. Set N_2/N_{RO} to 100%.

WARNING DO NOT EXCEED ENGINE OR HELICOPTER LIMITS.

8. Adjust collective lever as necessary to achieve an indicated torque of 95% or an indicated TOT of 810 °C, **whichever is reached first**. Adjust airspeed as necessary to maintain level flight.
9. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.

10. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N1 (%)
 - Engine Anti-icing (on or off)
 11. Set power lever of non-check engine to FLIGHT.
 12. Switch on generator of checked engine.
 13. Repeat above steps 5, through 12, to check other engine.
 14. Reset altimeter and, if necessary, switch on bleed air consumers.
 15. After flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-1A) to determine either maximum TOT or minimum required torque available under the prevailing ambient conditions.
 16. Power is assured and all performance data contained in this manual can be achieved when each engine's:
 - indicated TOT is equal to or less than the chart derived maximum TOT value; or
 - indicated torque is equal to or greater than the chart derived minimum required torque value;
- NOTE** If power is not assured on one or both engines then the total power available is less than the minimum specification and all performance data contained in this manual cannot be achieved. Refer to appropriate maintenance manual to determine cause of power loss.

EXAMPLE: (see Fig. 5-1)

Determine: Engine power assurance - Torque limit reached first

Known:

Indicated torque	95 %
Pressure altitude	4000 ft
OAT	-30 °C
Indicated TOT	730 °C
Anti-icing	OFF

- Solution:**
1. Enter chart at known torque (95%).
 2. Move vertically up to known pressure altitude (4000 ft) and then trace horizontally left through the TOT field.
 3. Reenter chart at known OAT (-30 °C) and move vertically up to intersect the above tracing.
 4. At point of intersection read maximum TOT = 740 °C.

Since the indicated TOT (730 °C) is lower than the obtained chart TOT limit (740 °C), the power assurance check for this engine is acceptable.

EXAMPLE: (see Fig. 5-1)

Determine: Engine power assurance - TOT limit reached first

Known:

Indicated TOT	810 °C
Pressure altitude	4000 ft
OAT	20 °C
Indicated torque	90 %
Anti-icing	OFF

- Solution:**
1. Enter chart at known OAT (20 °C).
 2. Move vertically up to known indicated TOT (810 °C) and trace horizontally right to intercept known PA (4000 ft).
 3. From this intercept point move vertically down to read minimum required torque = 86 %.

Since the indicated torque (90 %) is higher than the obtained chart minimum required torque (86 %), the power assurance check for this engine is acceptable.

INFLIGHT POWER CHECK

ALLISON 250-C20B

N_2 – 100%

N_1 – 95% OR ABOVE

GENERATOR – OFF

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

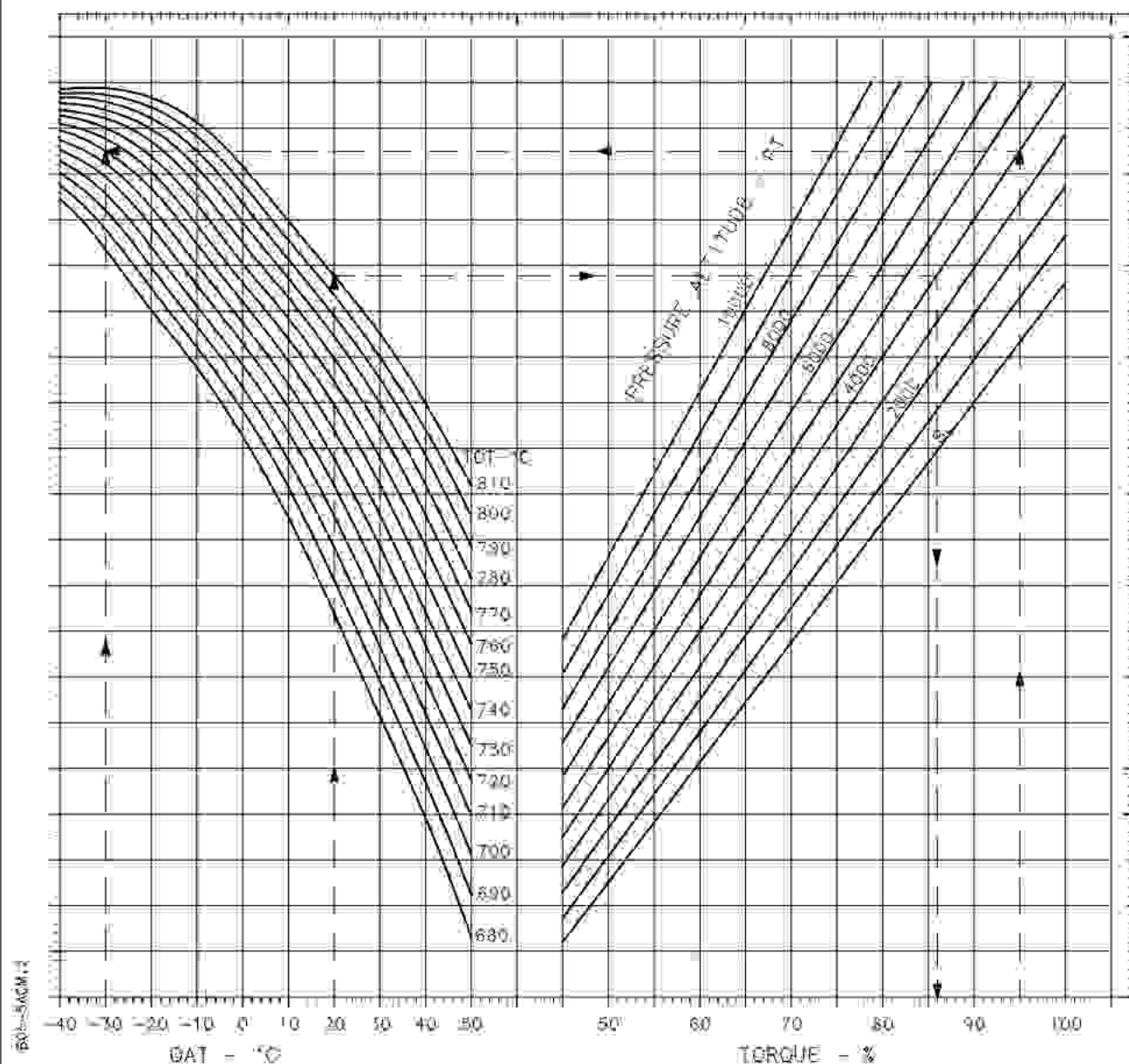


Fig. 5-1 Inflight Power Check (Anti-icing - OFF)

INFLIGHT POWER CHECK

ALLISON 250-C20B

N_2 – 100%

N_1 – 95% OR ABOVE

GENERATOR – OFF

ANTI-ICING – ON

BLEED AIR CONSUMERS – OFF

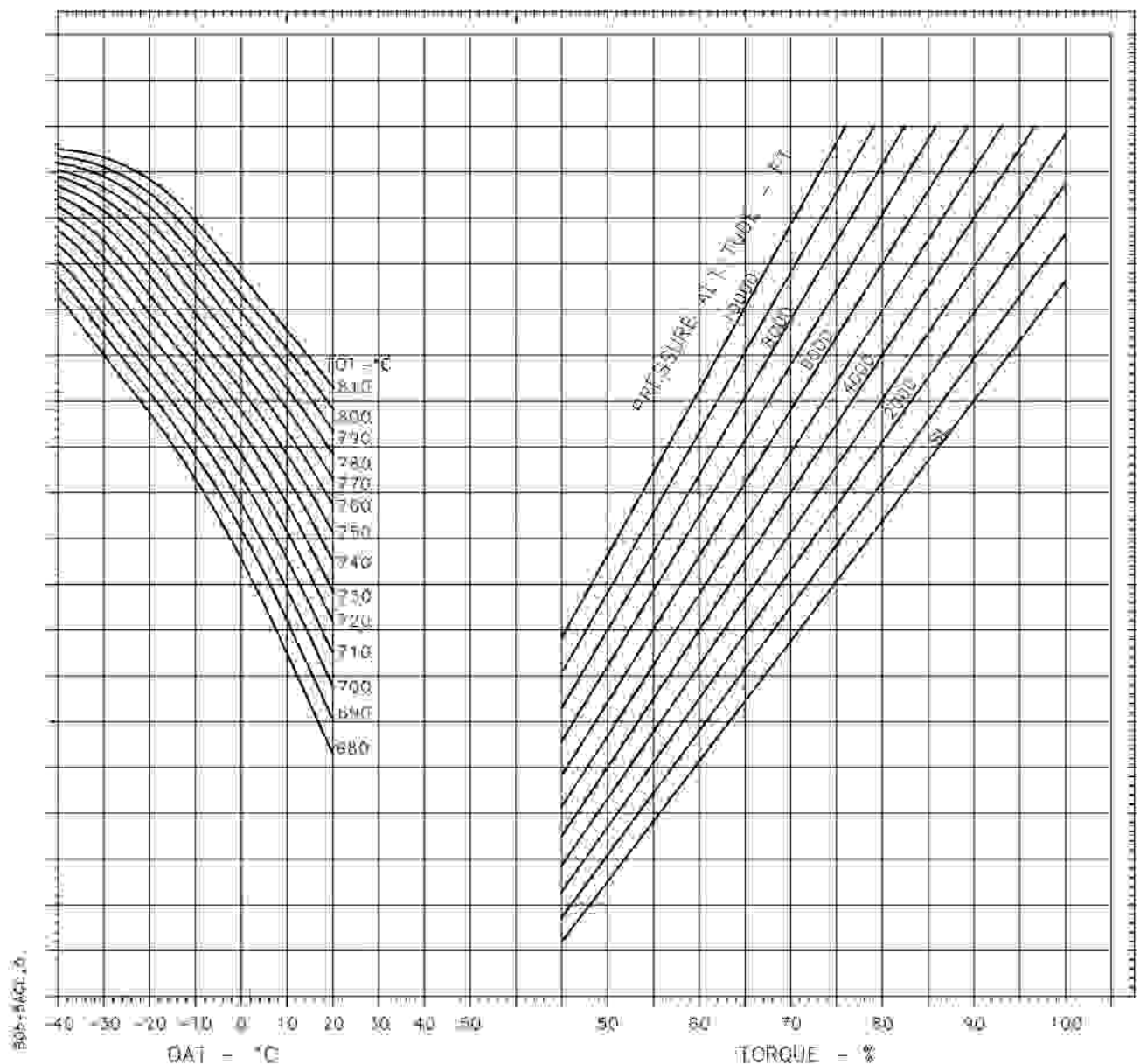


Fig. 5-1A. Inflight Power Check (Anti-icing - ON)

5.2.2 Engine Trend Check

NOTE The engine trend check is not an alternate procedure to the power assurance check. Only the results of the Power Assurance Check (see Para 5.2.1) shall be used to ensure that the engines are capable of developing the power necessary to meet the presented performance data.

The engine trend check is provided for the purpose of monitoring the long-term behaviour and recognizing sudden significant trend-line deviations of the engine performance at minimum operator expense. This check, which should be repeated on a regular basis based upon operator experience, is performed in cruise flight with both engines operating; each engine being checked separately, in turn. The non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).
2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y) or more.

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical consumption as much as possible.
5. Set N_2 at 100%.
6. Adjust collective lever as necessary to achieve an indicated torque of 86% or an indicated TOT of 779 °C, **whichever is reached first**.

NOTE If N_1 is not at least 95%, the engine bleed valve is open and the check result is not valid. In this case the check altitude should be increased.

7. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.
8. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N_1 (%)
 - Engine Anti-icing (on or off)
9. Repeat above steps 4. through 8. to check other engine.

10. Reset altimeter and, if necessary, switch on bleed air consumers.
11. Following flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-1A) to determine the minimum required torque available under the prevailing ambient conditions.

5.2.3 Power Trend Monitoring

A power trend monitoring shall be established to observe the deterioration of the engines with time in service as well as potential engine malfunctions.

NOTE The power trend check preconditions and the power trend check procedure have to be followed very accurately, otherwise the trend monitoring will not be reliable.

To establish a power trend chart, the values of Δ Torque (defined as measured torque minus chart derived minimum required torque) have to be recorded versus time in service (see Fig. 5-2).

CAUTION MAINTENANCE ACTION MUST BE CONSIDERED IF A Δ TORQUE VALUE DEVIATES SIGNIFICANTLY FROM THE ESTABLISHED TREND LINE.

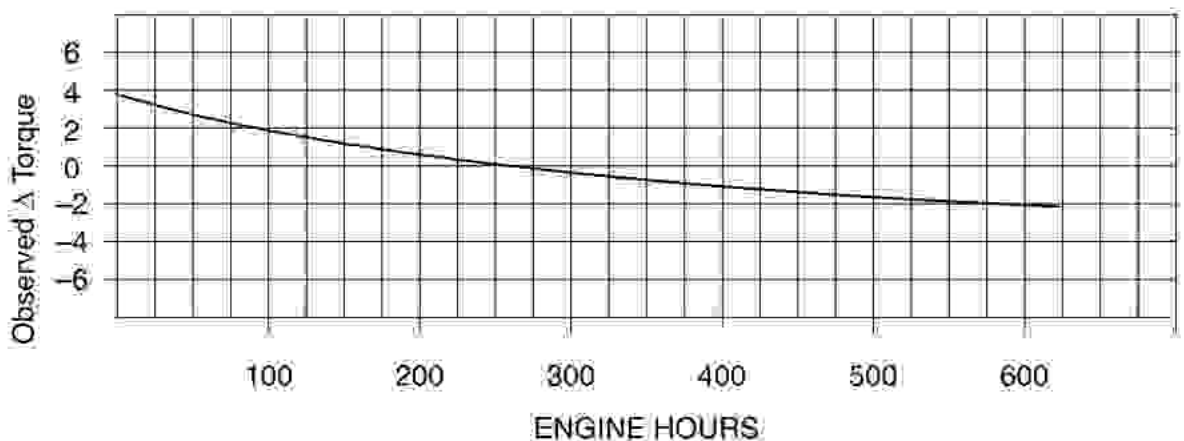


Fig. 5-2 Plotting Example for Power Trend Monitoring

5.3 DENSITY ALTITUDE

The density altitude chart (see Fig. 5-3) expresses density altitude in terms of pressure altitude and temperature. The less dense the air the higher the density altitude. For standard conditions of temperature and pressure, density altitude is identical to pressure altitude.

A high density altitude affects the performance of both rotors and engines. When density altitude is high, less lift is developed by the rotor blades for any given power setting than at standard conditions and the power output of the engines is reduced below the output for standard conditions.

Each takeoff and landing must be separately evaluated as density altitude may change considerably in a short period of time.

The true airspeed factor value $\frac{1}{\sqrt{\sigma}}$ is a conversion factor used to obtain true airspeed from calibrated airspeed by correcting for density altitude.

EXAMPLE (see Fig. 5-3)

Determine: **Density altitude (DA)**

True airspeed factor

True airspeed (TAS)

Known: OAT -14 °C
 Pressure altitude 5000 ft
 CAS 100 kt

- Solution:
1. Enter chart at known OAT (-14 °C).
 2. Move up to known pressure altitude (5000 ft).
 3. Move left and read density altitude = 2630 ft.
 4. Move right and read true airspeed factor = 1.04.
 5. Multiply the known calibrated airspeed (100 kt) by true airspeed factor ($\frac{1}{\sqrt{\sigma}} = 1.04$) to obtain true airspeed.

$$TAS = CAS \times \frac{1}{\sqrt{\sigma}} = 100 \times 1.04 = \underline{104 \text{ KTAS}}$$

DENSITY ALTITUDE CHART

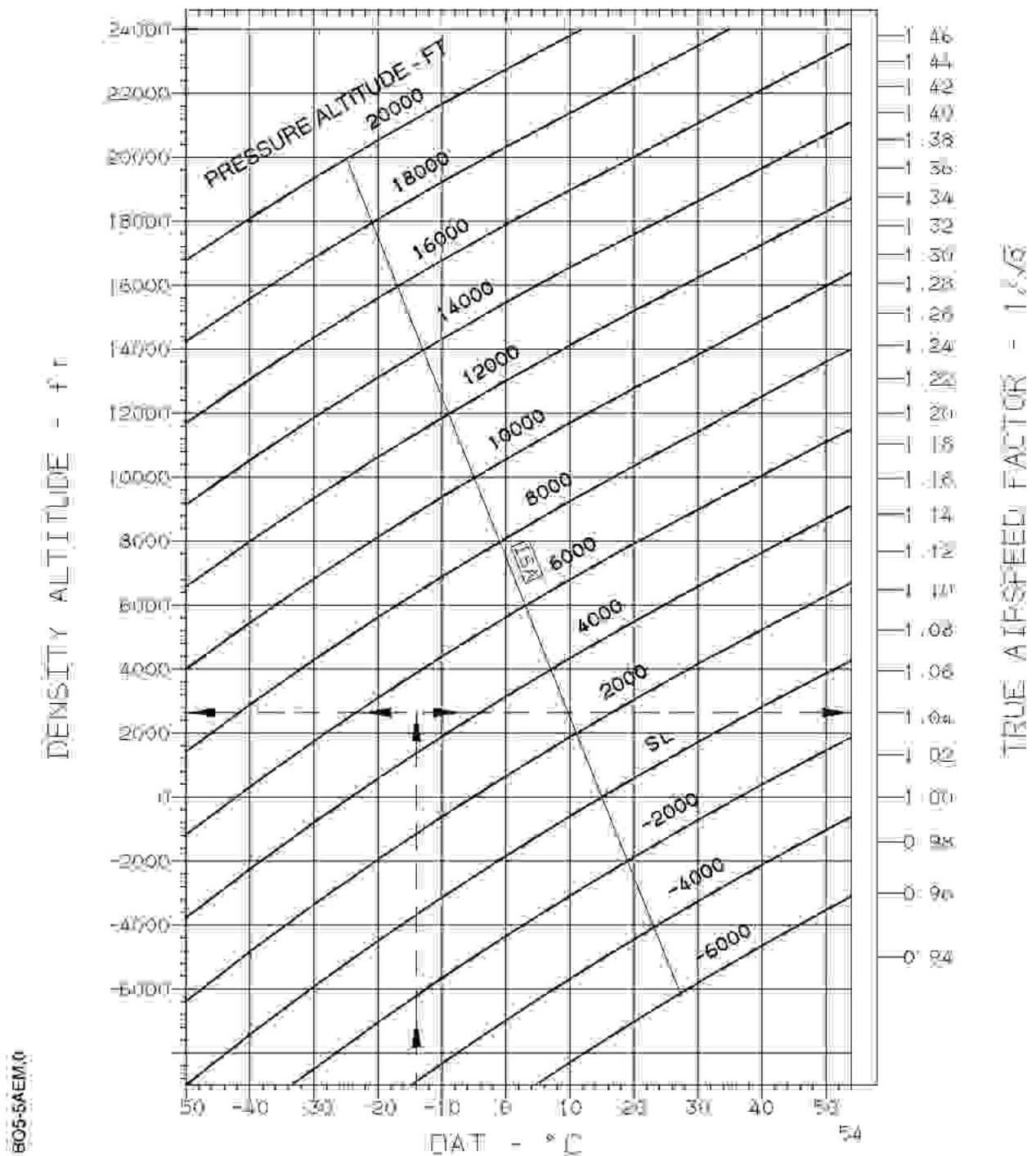


Fig. 5-3 Density Altitude Chart

5.4 AIRSPEED CALIBRATION

The airspeed calibration chart (see Fig. 5-4) is provided to determine Calibrated Airspeed (CAS) from Indicated Airspeed (IAS) using the pitot static system installation-error correction for level flight.

NOTE Indicated airspeed assumes zero instrument error.

The corrections apply for all practical altitudes and gross masses, and with the helicopter flown with no appreciable yaw.

EXAMPLE: (see Fig. 5-4)

Determine: **Calibrated airspeed (CAS), level flight**

Known: Indicated airspeed 90 KIAS

Solution: 1. Enter chart at known indicated airspeed (90 KIAS).
2. Move up to LEVEL FLIGHT line.
3. Move left and read calibrated airspeed = 91 KCAS.

AIRSPEED CALIBRATION CHART

CLEAN CONFIGURATION

NOTE INDICATED AIRSPEED ASSUMES ZERO INSTRUMENT ERROR

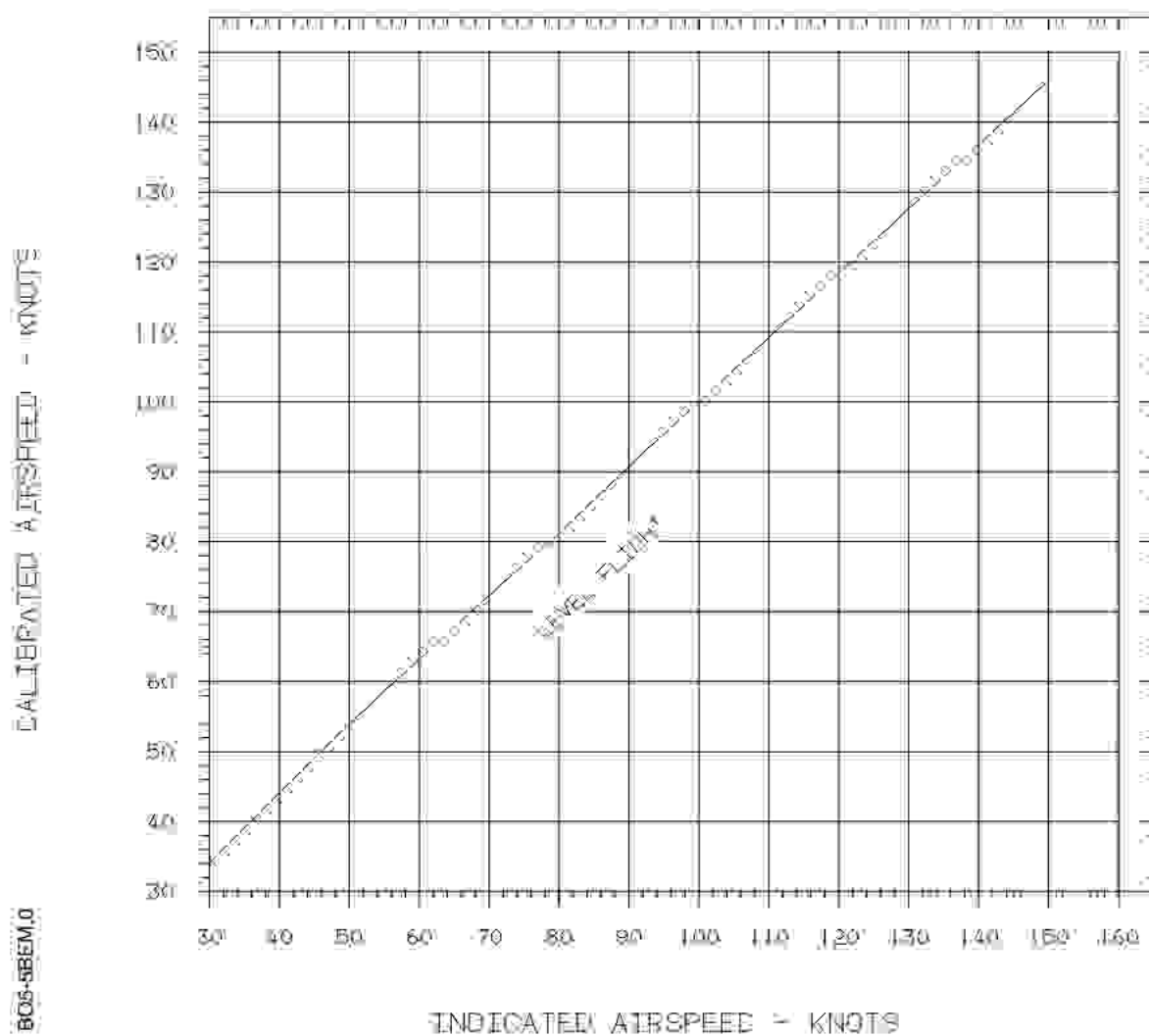


Fig. 5-4 Airspeed Calibration Chart

5.5 HEIGHT-VELOCITY DIAGRAMS (see Fig. 5-5 and 5-6)

CAUTION AVOID OPERATION WITHIN THE CAUTION AREA - POWER LOSS OR ENGINE FAILURE WHILE OPERATING WITHIN THIS AREA IS LIKELY TO RESULT IN DAMAGE TO THE HELICOPTER.

The operator must become familiar with the height-velocity diagrams to determine what altitudes and airspeeds are required to safely make an emergency landing in case of a power loss or engine failure. The altitude-airspeed combinations where a safe emergency landing on a smooth level firm surface may not be possible are represented by the caution (cross-hatched) area of the diagrams.

Two groups of height-velocity diagrams are presented: the first for **single engine failure** represented in varying gross mass and pressure altitude, and the second for **double engine failure** also represented in varying gross mass and pressure altitude.

HEIGHT-VELOCITY DIAGRAMS

SINGLE ENGINE FAILURE

ISA CONDITIONS

ANTI-ICING - OFF

BLEED AIR CONSUMERS - OFF

CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

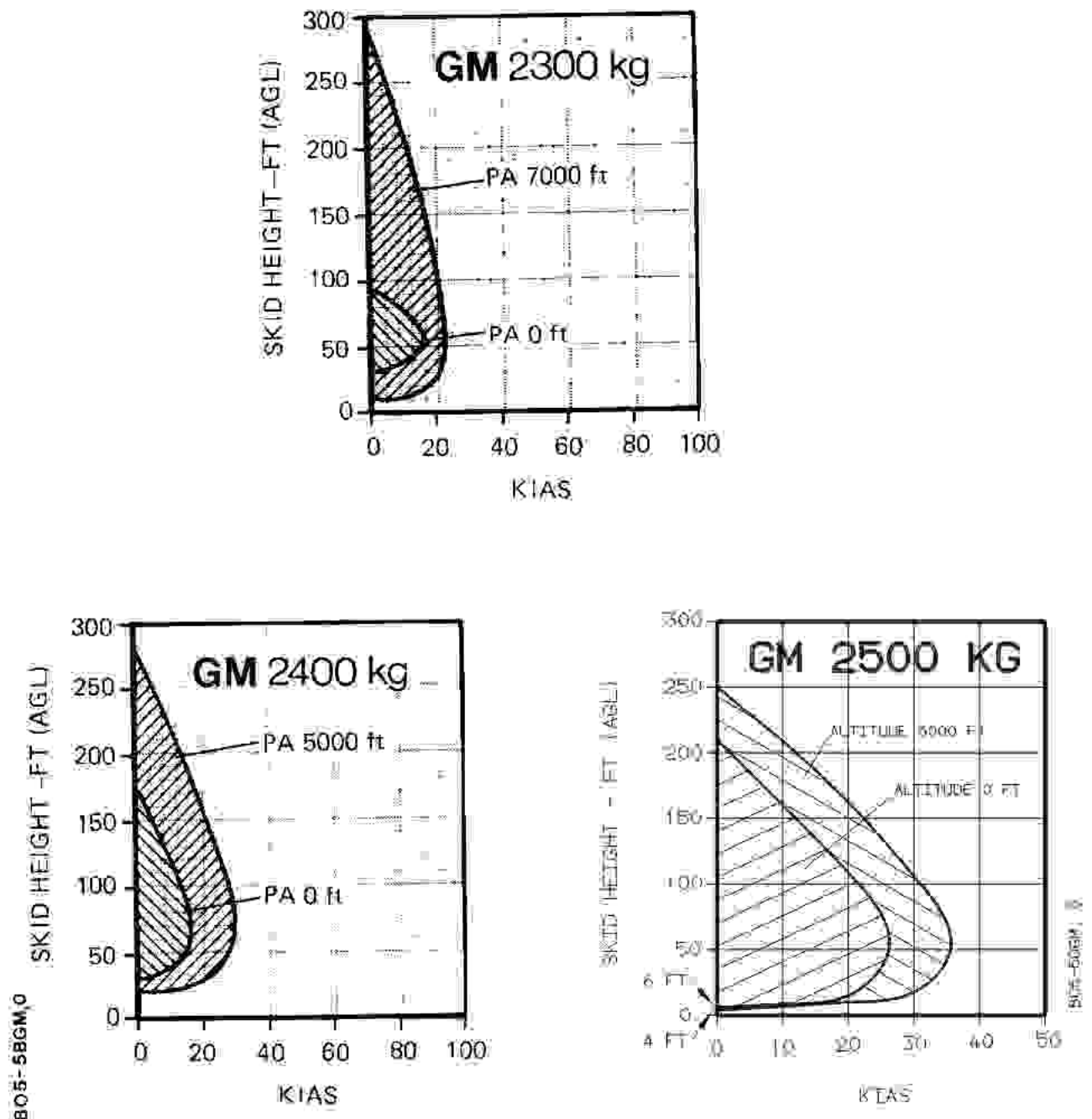


Fig. 5-5 Height-velocity Diagrams - Single Engine Failure

HEIGHT-VELOCITY DIAGRAMS

DOUBLE ENGINE FAILURE

ISA CONDITIONS

CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

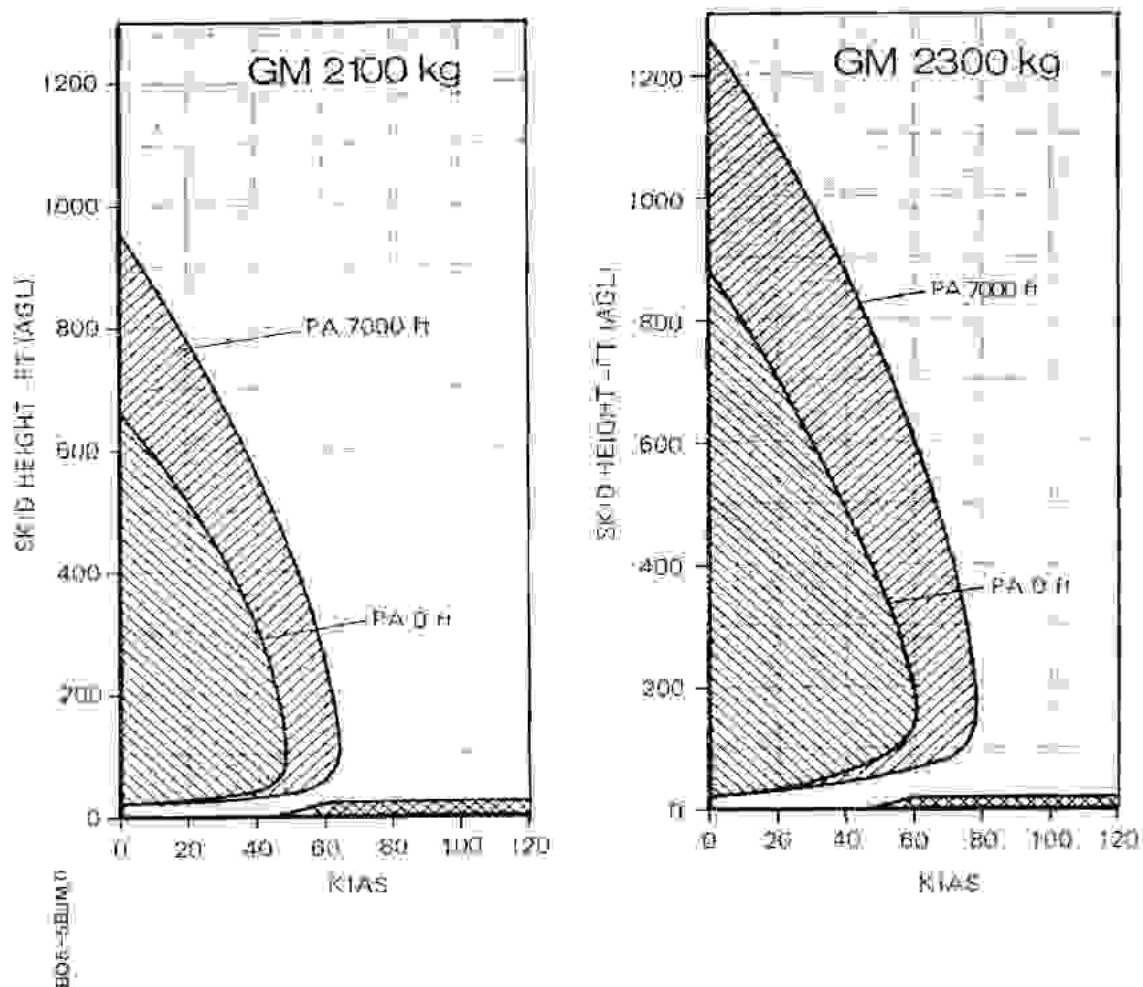


Fig. 5-6 Height-velocity Diagrams - Double Engine Failure (Sheet 1 of 2)

HEIGHT-VELOCITY DIAGRAMS

DOUBLE ENGINE FAILURE

ISA CONDITIONS

CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

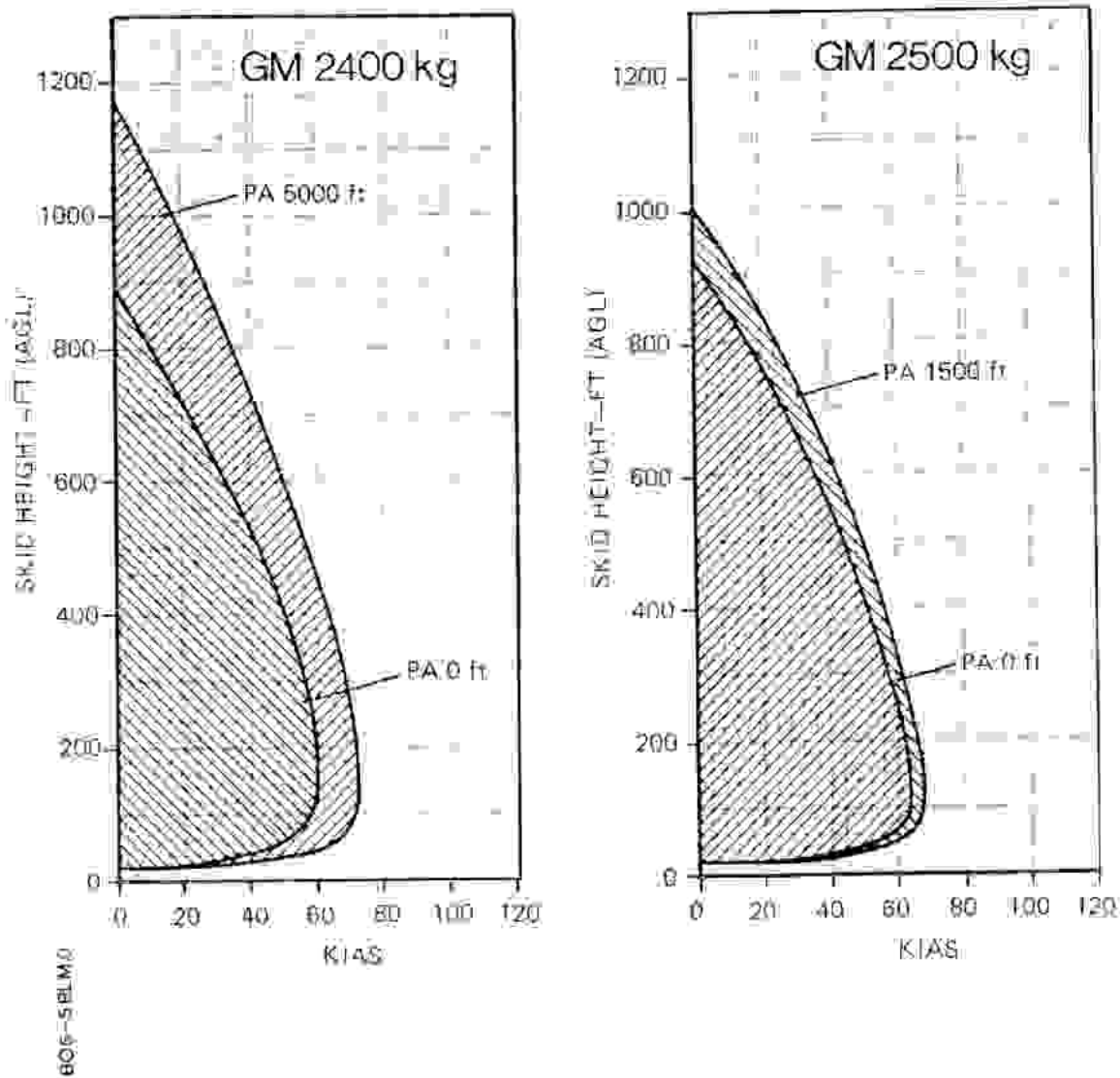


Fig. 5-6 Height-velocity Diagrams - Double Engine Failure (Sheet 2 of 2)

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5.6. HOVER CEILING (see Fig. 5-7 thru 5-10)

These hover ceiling charts present the maximum gross mass for hover in and out of ground effect at various conditions of pressure altitude and outside air temperature, and under different power settings. The charts for hover in ground effect are based on a skid height of 3 ft and are calculated for hovering with engine anti-icing OFF.

NOTE • For operations requiring engine anti-icing ON, reduce the chart derived gross mass value by 120 kg.

- OGE hover ceiling diagrams are valid for no wind or headwind.

EXAMPLE: (see Fig. 5-7)

Determine: Maximum gross mass for hovering, all engines operating

Known: OAT 8 °C
 Pressure altitude 8000 ft

Solution: 1. Enter chart at known OAT (8 °C).
 2. Move vertically upward to known pressure altitude (8000 ft).
 3. Move horizontally left and read maximum gross mass = 2265 kg.

Determine: Maximum gross mass for hovering, all engines operating and engine anti-icing ON

Known: OAT 8 °C
 Pressure altitude 8000 ft

Solution: 1. Enter chart at known OAT (8 °C).
 2. Move vertically upward to known pressure altitude (8000 ft).
 3. Move horizontally left and read maximum gross mass = 2265 kg.
 4. Subtract 120 kg from chart derived value (2265 kg) for 2145 kg maximum gross mass for hovering with all engines operating and engine anti-icing ON.

HOVER CEILING IN GROUND EFFECT

2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

SKID HEIGHT – 3 FT

BLEED AIR CONSUMERS – OFF

NOTE: FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE GROSS MASS LIMIT LINE SHALL APPLY.

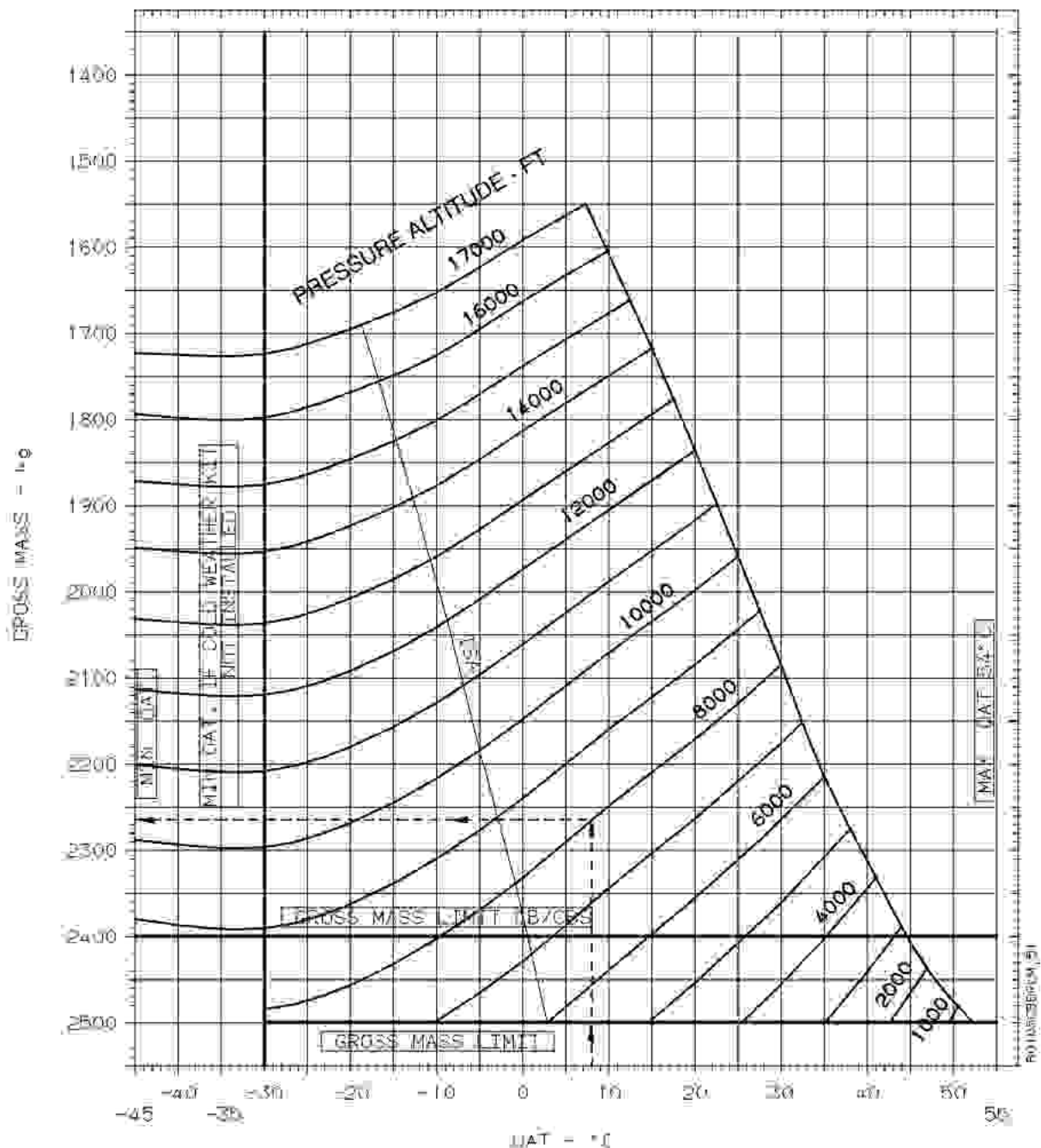


Fig. 5-7 Hover Ceiling I G E - TOP

HOVER CEILING IN GROUND EFFECT

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

SKID HEIGHT – 3 FT

BLEED AIR CONSUMERS – OFF

NOTE FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE GROSS MASS LIMIT LINE SHALL APPLY.

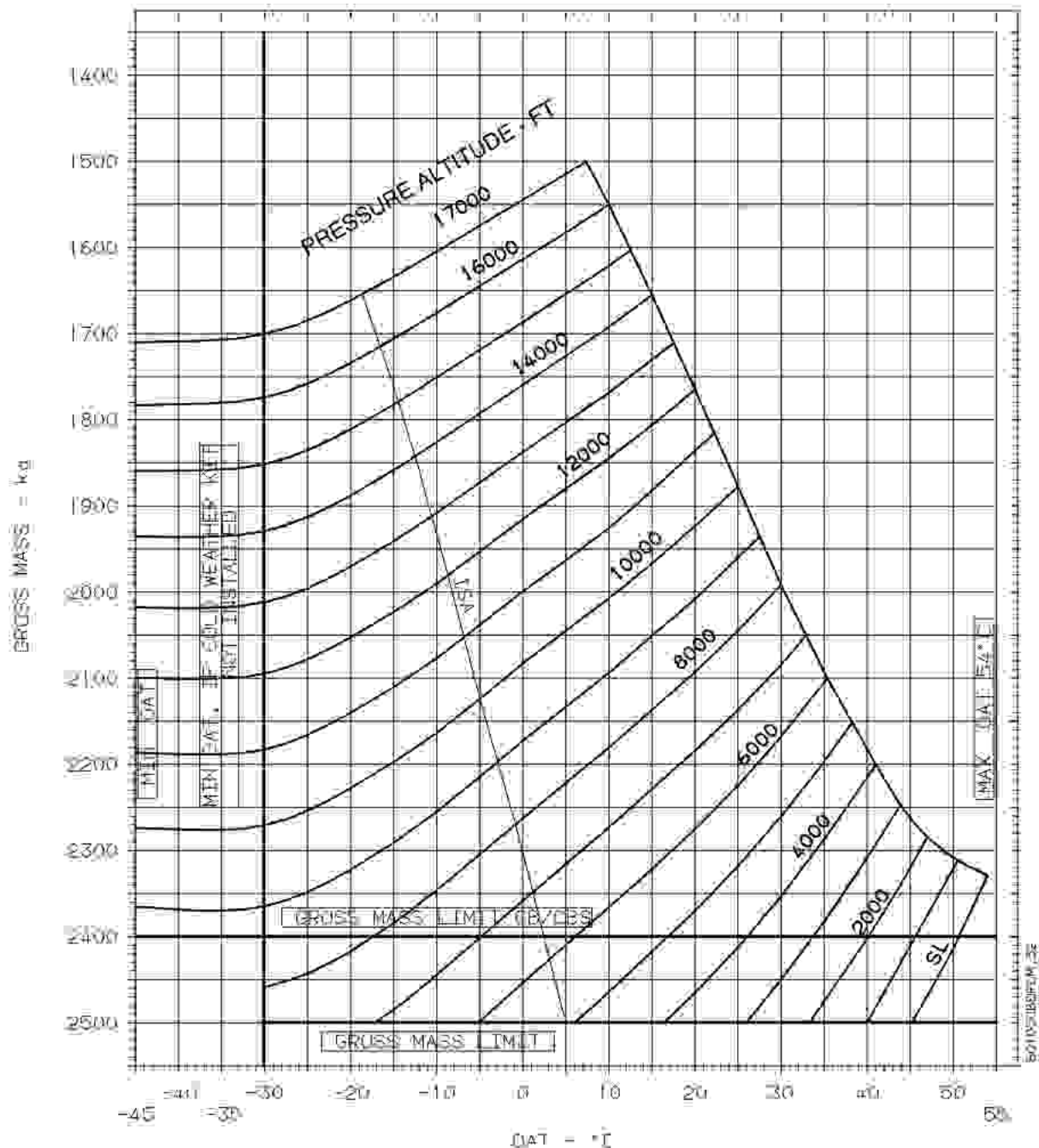


Fig. 5-8 Hover Ceiling I G E - MCP

HOVER CEILING OUT OF GROUND EFFECT

2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING - OFF

BLEED AIR CONSUMERS - OFF

NOTE: FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE GROSS MASS LIMIT LINE SHALL APPLY.

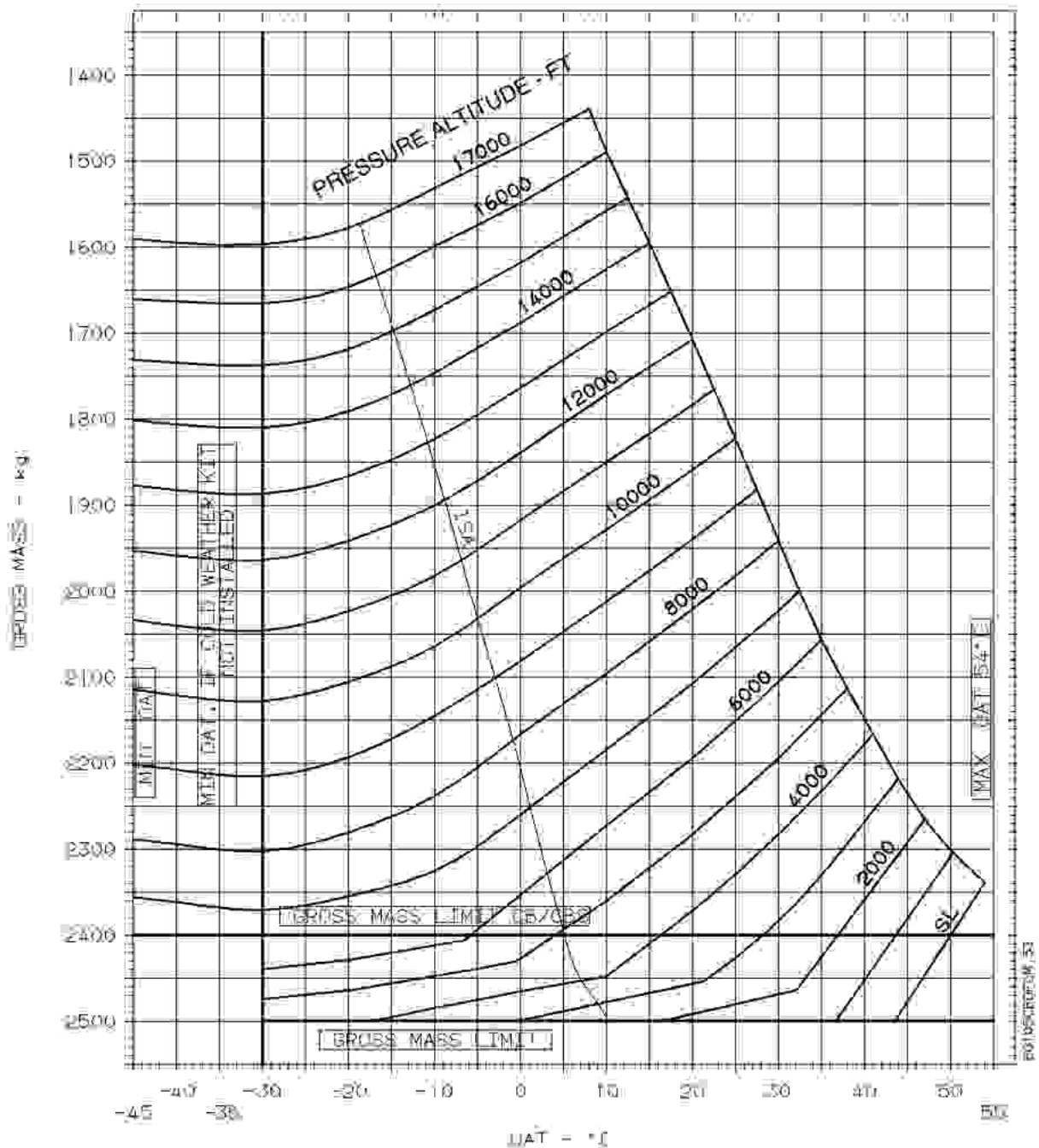


Fig. 5-9 Hover Ceiling O G E- TOP

HOVER CEILING OUT OF GROUND EFFECT

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

NOTE FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE GROSS MASS LIMIT LINE SHALL APPLY.

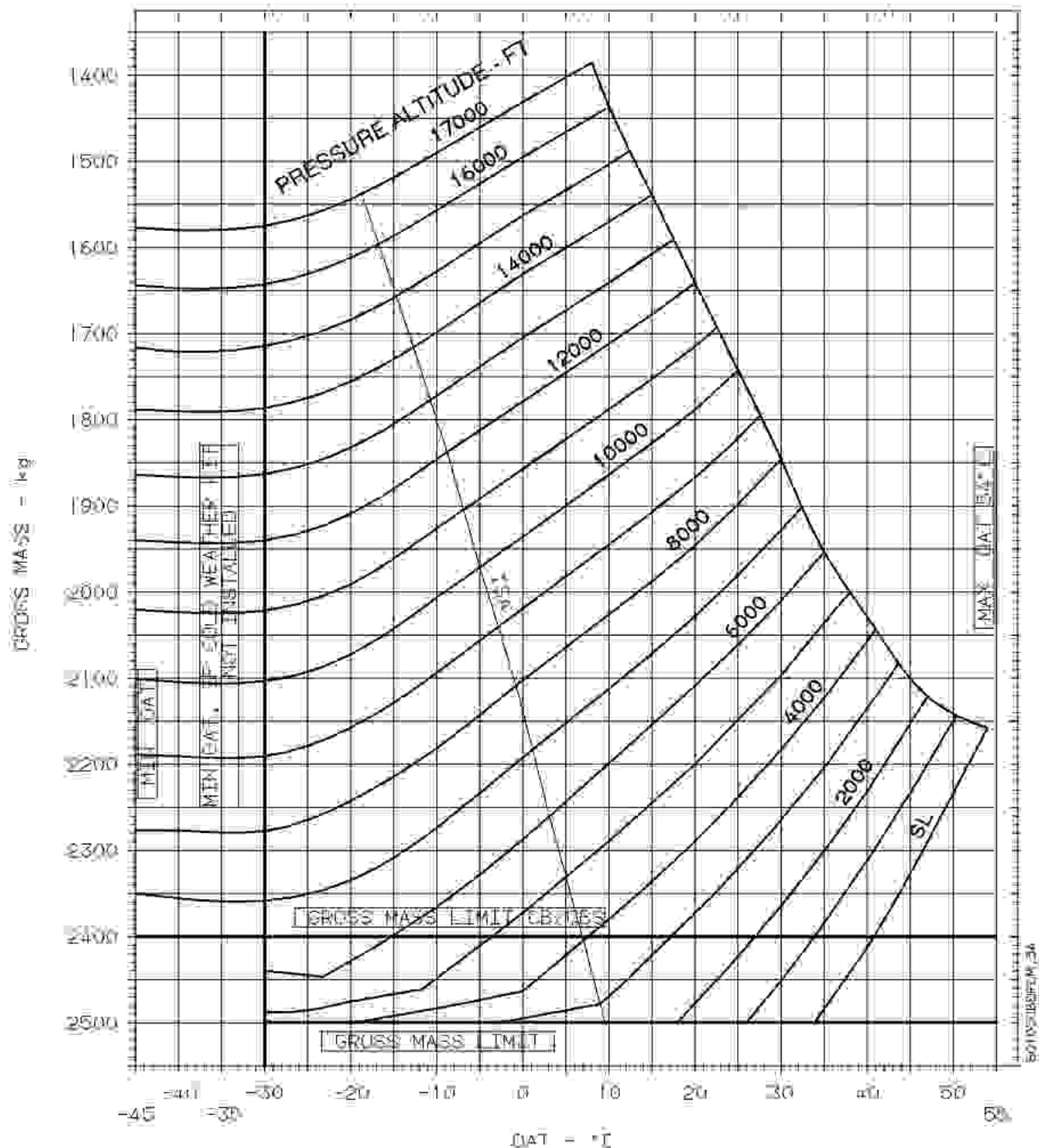


Fig. 5-10 Hover Ceiling O G E - MCP

5.7 RATE OF CLIMB

The rate-of-climb charts are presented in two groups: the first group for operations with all engines operating (AEO, Fig. 5-11 and 5-12) and the second group for operations with one engine inoperative (OEI, Fig. 5-13 and 5-14). Climb rates are functions of power, pressure altitude, outside air temperature, and gross mass.

NOTE • Chart data are based on the clean configuration:

- Chart derived rate-of-climb values are established using the best rate-of-climb speed (V_Y).
- For AEO operations requiring engine anti-icing ON, reduce the chart derived rate-of-climb value by 300 fpm.
- For OEI operations requiring engine anti-icing ON, reduce chart derived rate-of-climb value by 200 fpm.

EFFECTIVITY Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg

AEO Rate of Climb (Fig. 5-11 and 5-12)

1. Reduce the known gross mass by 100 kg.
2. Determine the rate-of-climb value using this corrected gross mass.
3. To obtain final rate-of-climb value for AEO operations, reduce the chart derived rate-of-climb value by 200 fpm.

OEI Rate of Climb (Fig. 5-13 and 5-14)

To obtain final rate-of-climb value for OEI operations, reduce chart derived rate-of-climb value by 100 fpm.

EFFECTIVITY ALL

EXAMPLE: (see Fig. 5-11)

Determine: **Rate of climb**

Known: Pressure altitude 11000 ft
OAT 20 °C
Gross mass 1820 kg

- Solution:**
1. Enter chart at known pressure altitude (11000 ft).
 2. Move up to known OAT (20 °C).
 3. Move right to reference line and then down following curve of guide lines until intercepting known gross mass (1820 kg).
 4. From this intercept point, move right and read rate of climb = 1420 fpm.

RATE OF CLIMB
2 X ALLISON 250-C20B

AEO — TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING — OFF

BEST RATE-OF-CLIMB SPEED — 60 KIAS

BLEED AIR CONSUMERS — OFF

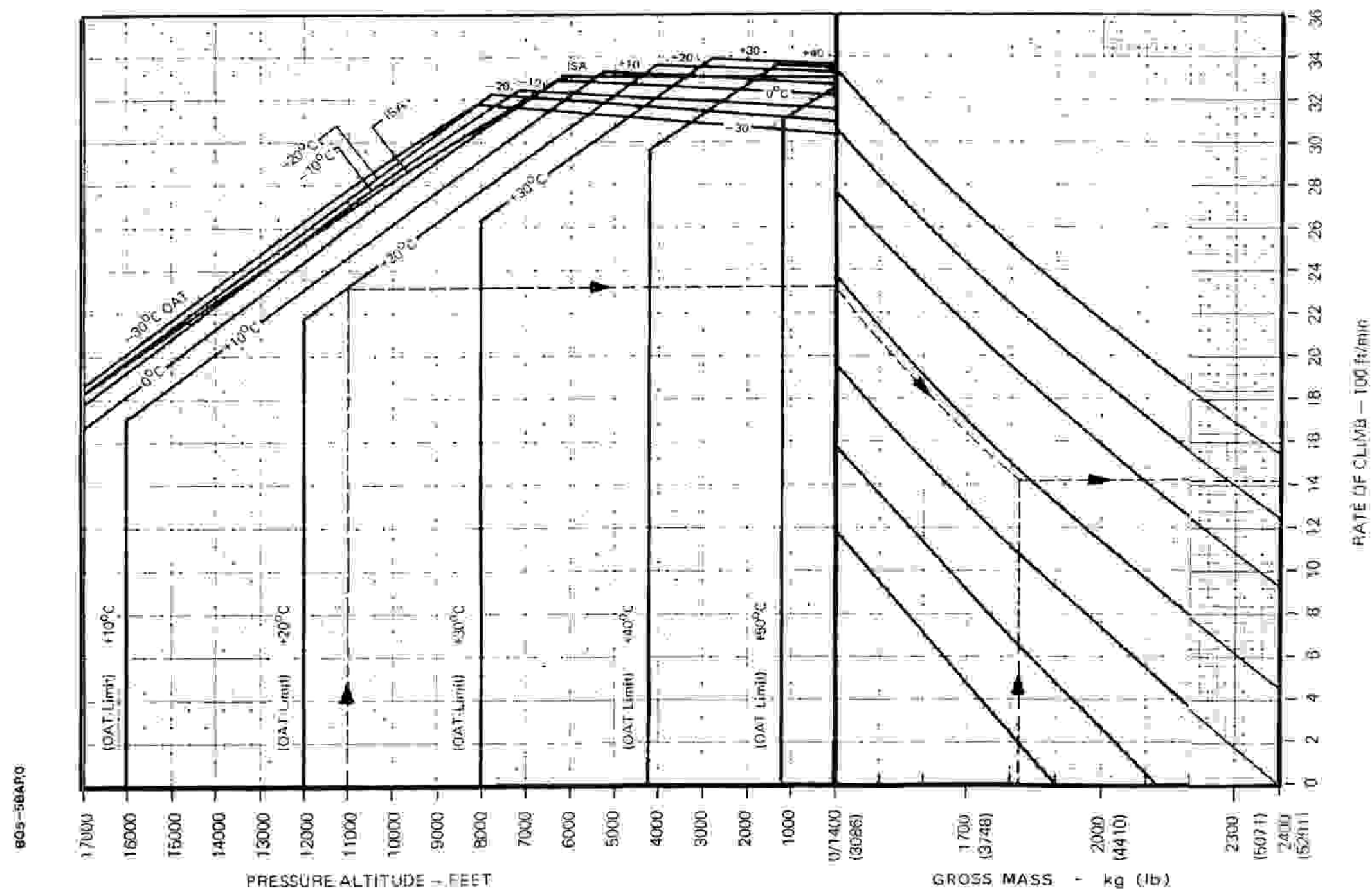


Fig. 5-11 Rate of Climb - AEO, TOP (Sheet 1 of 2)

RATE OF CLIMB

EXTREME LOW TEMPERATURES (-30 TO -45 °C)

2 X ALLISON 250-C20B

AEO - TAKEOFF POWER (810 °C TOT, 86% TORQUE)

BEST RATE-OF-CLIMB SPEED - 60 KIAS

ANTI-ICING - OFF

BLEED AIR CONSUMERS - OFF

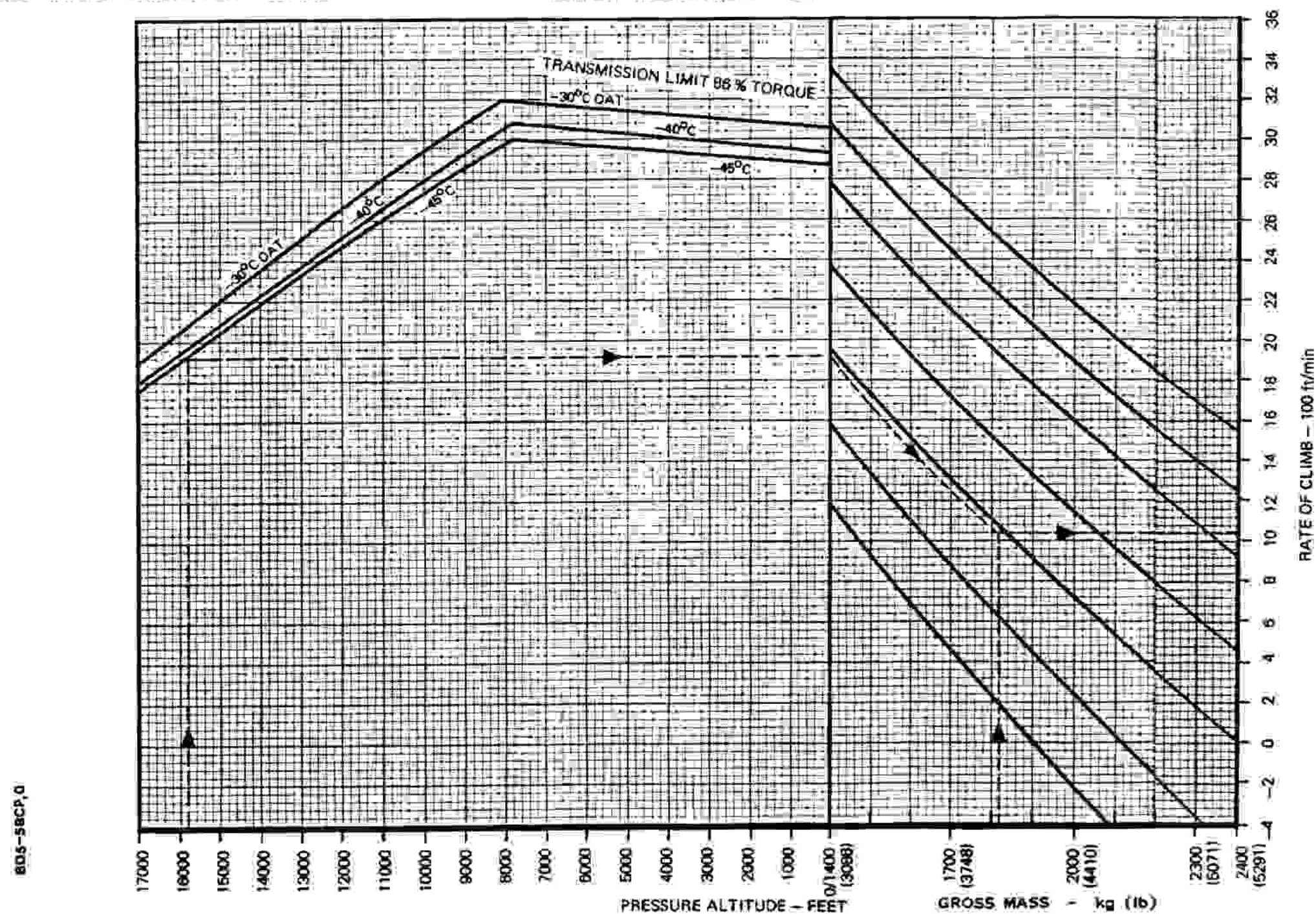


Fig. 5-11 Rate of Climb - AEO, TOP (Sheet 2 of 2)

RATE OF CLIMB

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

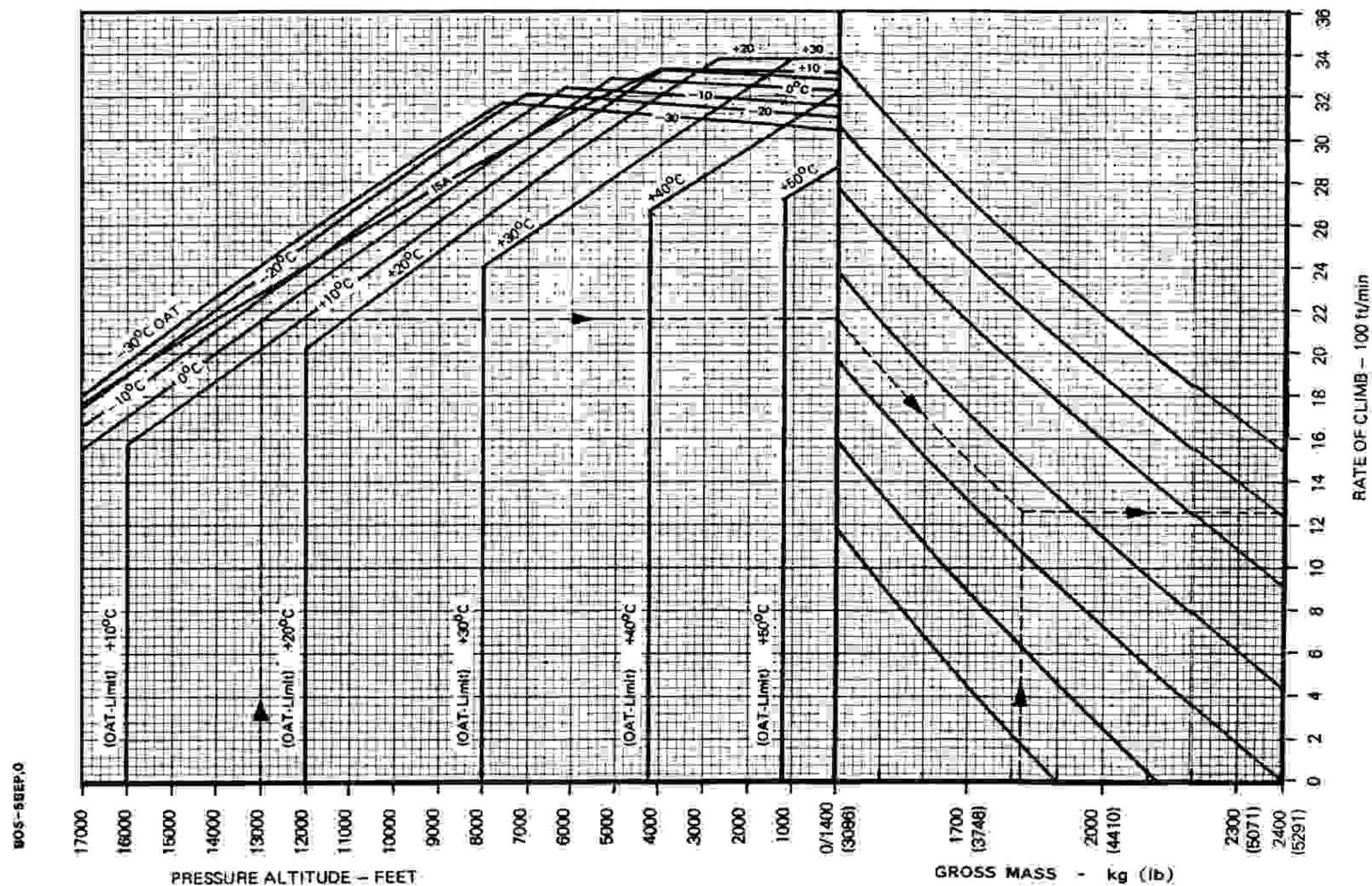


Fig. 5-12 Rate of Climb - AEO, MCP (Sheet 1 of 2)

RATE OF CLIMB

EXTREME LOW TEMPERATURES (-30 TO -45 °C)
2 X ALLISON 250-C20B

AEO - MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING - OFF

BEST RATE-OF-CLIMB SPEED - 60 KIAS

BLEED AIR CONSUMERS - OFF

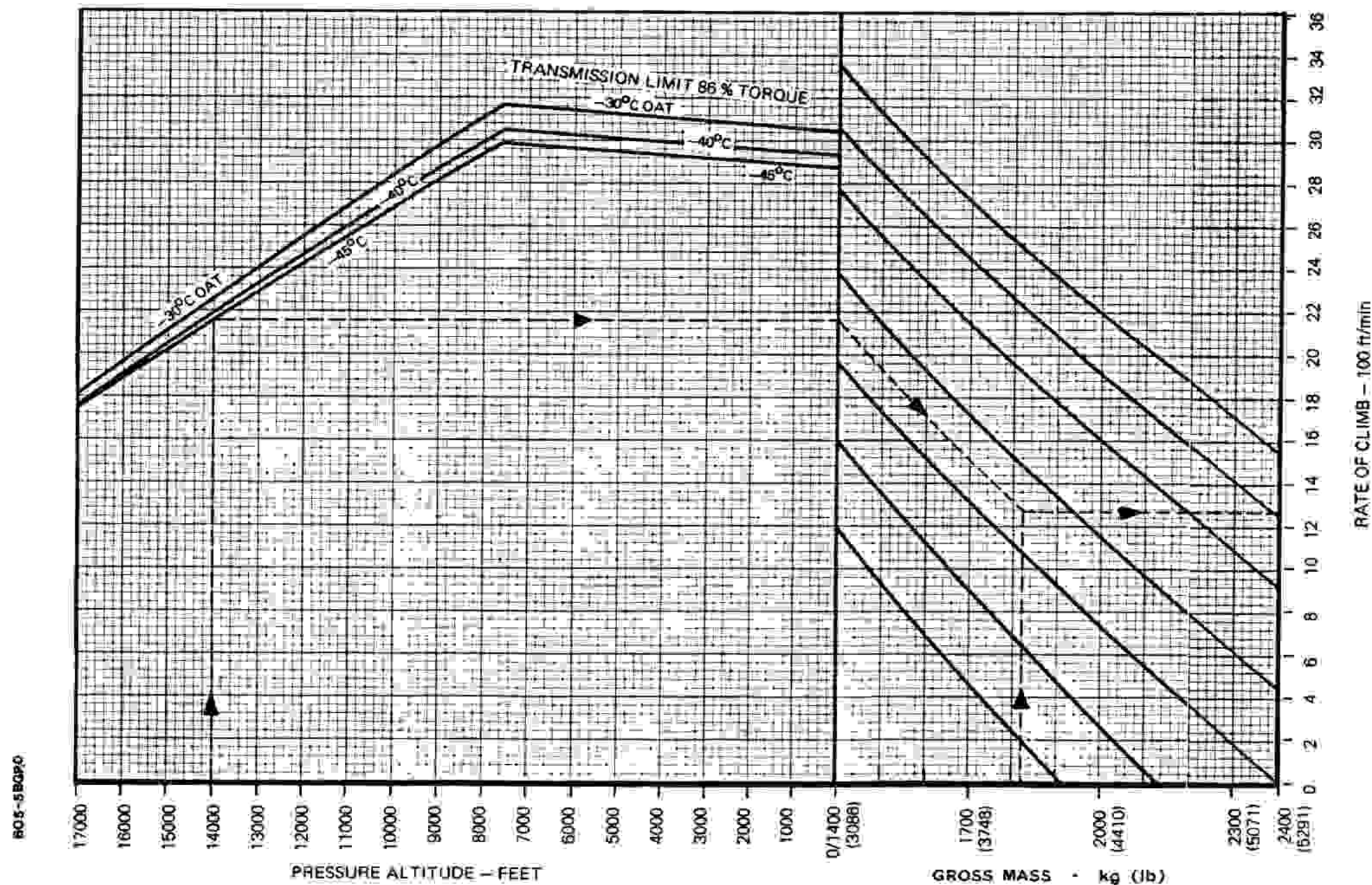


Fig. 5-12 Rate of Climb - AEO, MCP (Sheet 2 of 2)

RATE OF CLIMB 1 X ALLISON 250-C20B



OEI – EMERGENCY POWER (810 °C TOT, 95% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

NOTE: WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

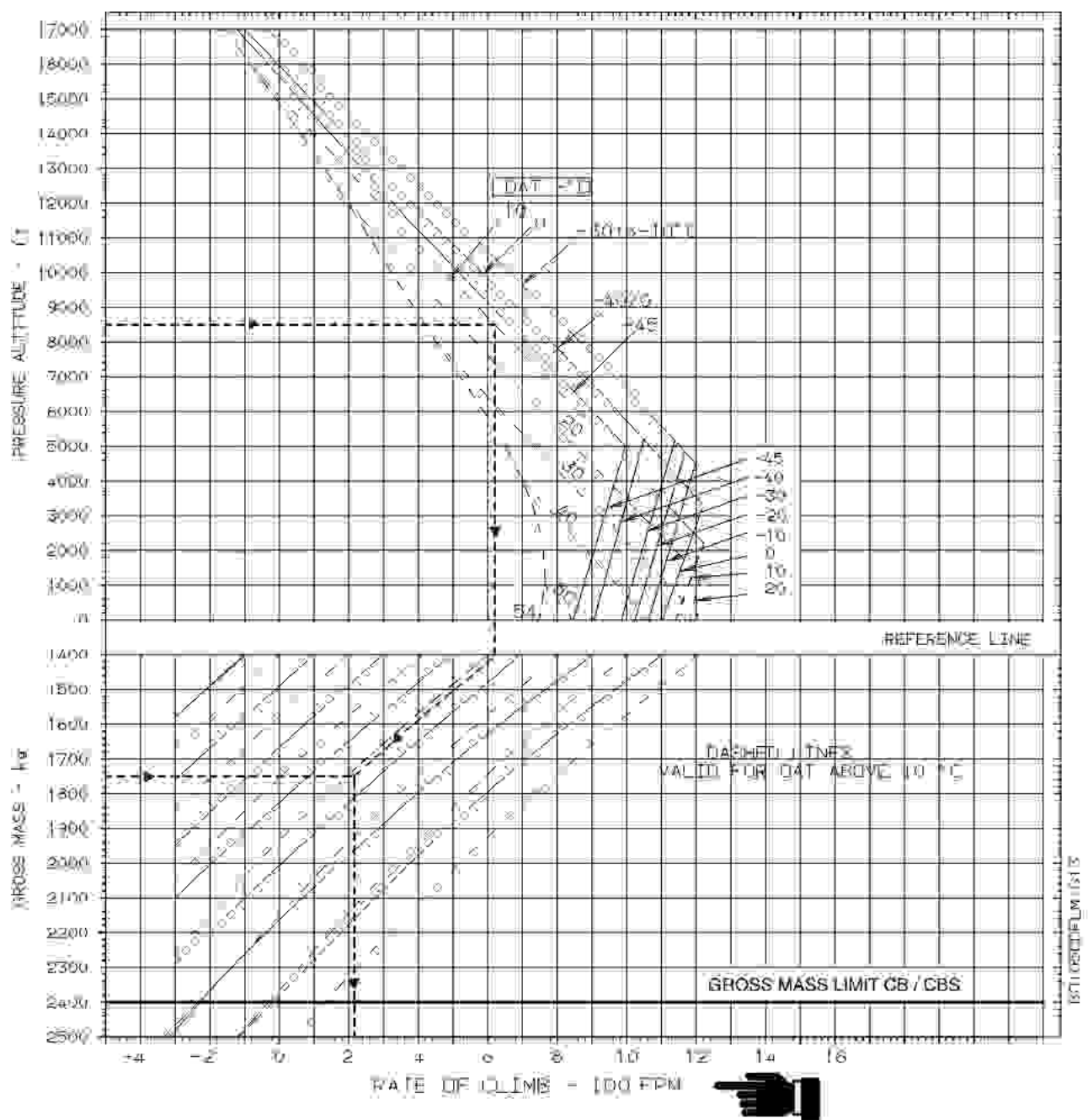


Fig. 5-13. Rate of Climb- OEI, EPWR

RATE OF CLIMB 1 X ALLISON 250-C20B



OEI = MAX CONTINUOUS POWER (779 °C TOT, 95% TORQUE)

ANTI-ICING = OFF

BEST RATE-OF-CLIMB SPEED = 60 KIAS

BLEED AIR CONSUMERS = OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

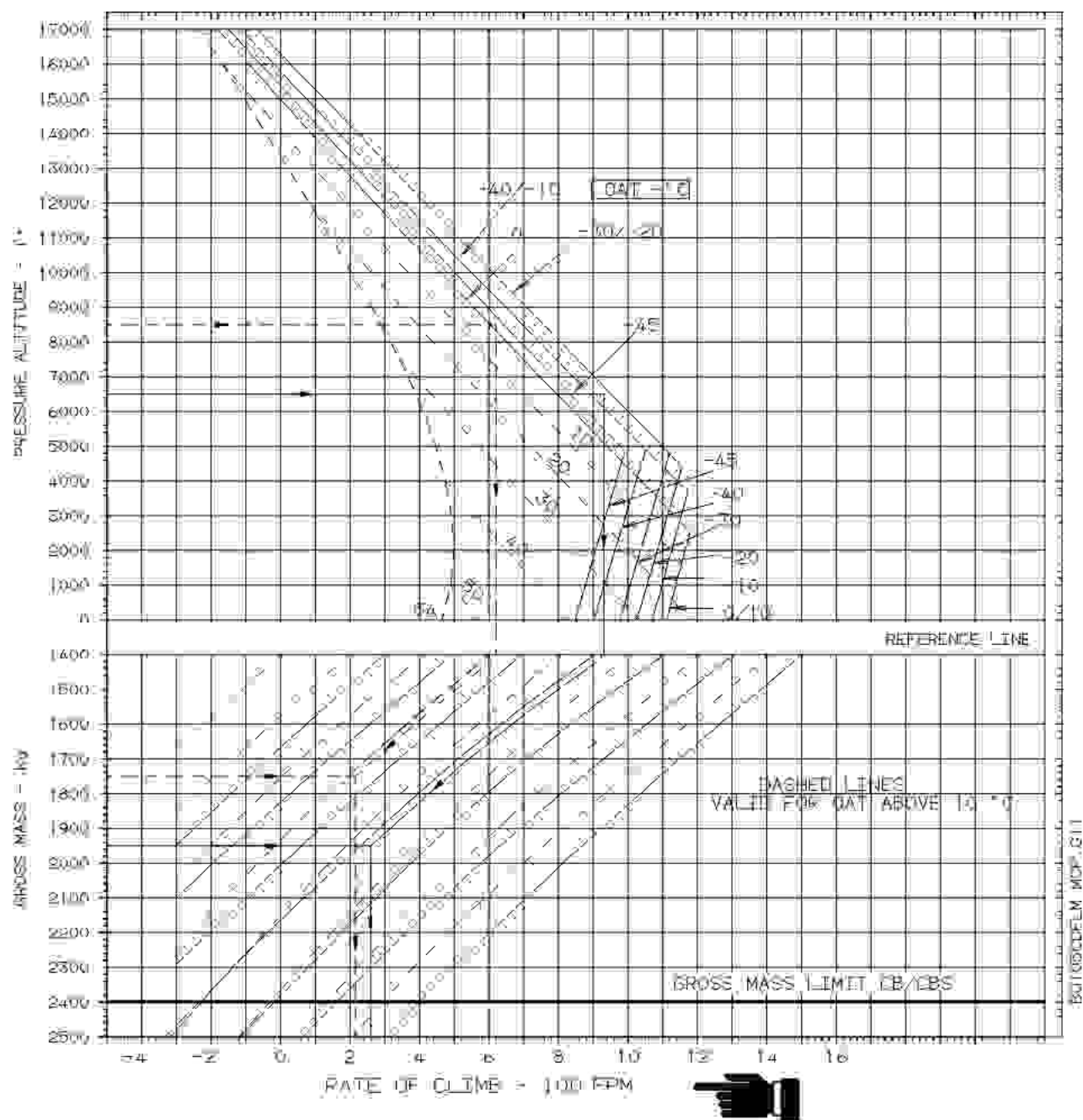


Fig. 5-14 Rate of Climb- OEI, MCP

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5.8 CLEAR AIRFIELD LANDING DISTANCE

Use of the clear airfield landing profile (see Fig. 5-15) assures the capability to land safely (on a smooth level surface) should an engine failure occur any time prior to or during an approach.

NOTE Go-around capability is NOT assured for OEI operation when using this clear airfield landing profile.

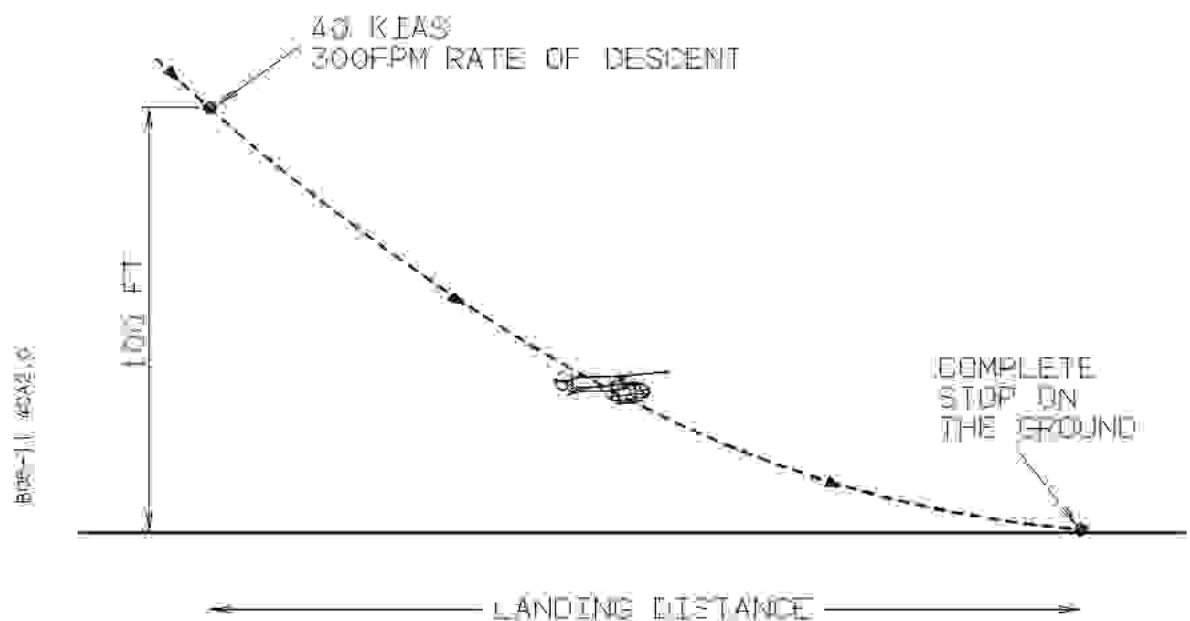


Fig. 5-15 Landing Profile - Clear Airfield

The landing distance chart (see Fig. 5-16) provides single engine landing performance data for various combinations of pressure altitude and outside air temperature. The chart shows the landing distance required for the helicopter to come to a complete stop on a smooth, hard and dry level surface. The presented data are valid for all gross masses up to 2500 kg.

To achieve the values derived from the landing distance chart, the landing approach path at a 100-ft height should be established with OEI power required to maintain a 300-fpm rate of descent at 40 KIAS.

NOTE For operations requiring engine anti-icing ON, increase the chart derived landing distance value by 100 ft.

EXAMPLE: (see Fig. 5-16)

Determine: **Landing Distance Required to Clear a 100-ft Obstacle**

Known: OAT 13 °C
Pressure altitude 2000 ft

Solution:

1. Enter chart at known OAT (13 °C).
2. Move horizontally right to known pressure altitude reference line (2000 ft).
3. From this intersection point move vertically down and read landing distance = 1805 ft.

CLEAR AIRFIELD LANDING DISTANCE
(FROM 100-FT HEIGHT TO A COMPLETE STOP ON GROUND)
 1 X ALLISON 250-C20B

POWER AS REQUIRED

ANTI-ICING - OFF

40 KIAS AND 300 FPM RATE OF DESCENT AT 100 FT HEIGHT

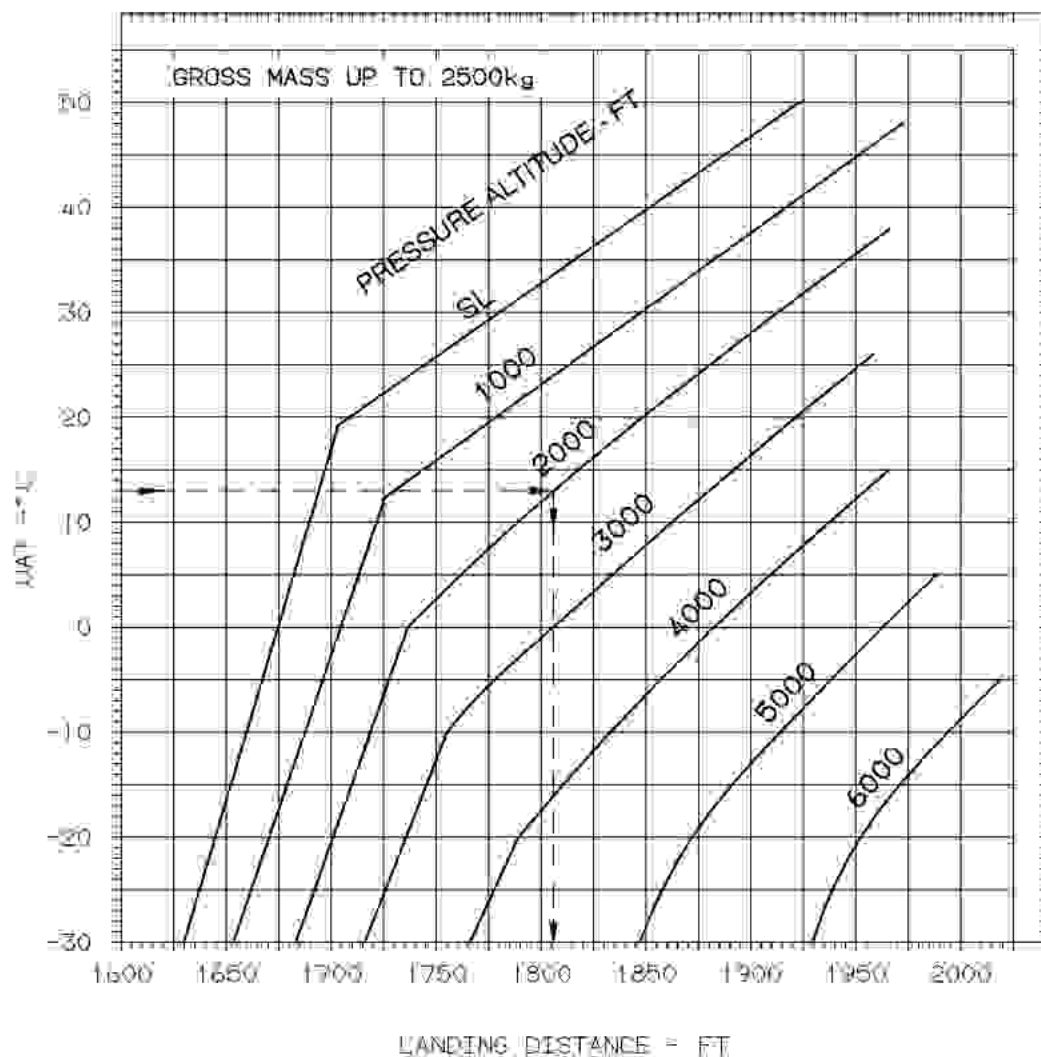
NOTE WITH ENGINE ANTI-ICING ON THE LANDING DISTANCE IS INCREASED BY 100 FT

Fig. 5-16 Landing Distance - Clear Airfield

